Deep Learning Summer School 2026

on AI for Economics and Finance

Reading List

Torino, Italy · August 24–26, 2026

Before the Summer School (Pre-Reading & Pre-Coding)

Everyone should complete these two items before arriving in Torino.

- **Coding — neural networks for economic models.** Simon Scheidegger, tutorials on using neural networks to solve simple Brock–Mirman models (Jupyter notebooks).

  Read and run both notebooks: Notebook 1 — deterministic Brock–Mirman, and Notebook 2 — stochastic Brock–Mirman. New to Python? Work through the Python refresher in this repository first, together with the introduction to Jupyter notebooks.

- **Reading — mathematical background.** Marc Peter Deisenroth, A. Aldo Faisal & Cheng Soon Ong, *Mathematics for Machine Learning*, Cambridge University Press, 2020 (full text free online). [link]

  Skim **Chapters 2–5** (linear algebra, analytic geometry, matrix decompositions, vector calculus) and **Chapter 6** (probability). This is the level of fluency the lectures assume.

Recommended background

- Goodfellow, Bengio & Courville (2016). *Deep Learning.* MIT Press (full text free online). [link]

  The canonical deep-learning textbook. Skim Part I and Chapters 6–8 (feedforward networks, regularization, optimization/SGD). Chapter 10 (sequence modelling with recurrent nets) is useful preparation for Days 2 and 3.

Day 1 — Deep Equilibrium Nets, Deep Surrogates, and Heterogeneous Agents

Simon Scheidegger (University of Lausanne), Yucheng Yang (University of Zurich)

Introduction to Deep Equilibrium Nets, and hands-on exercises (morning)

- Azinovic, Gaegauf & Scheidegger (2022). *Deep Equilibrium Nets.* International Economic Review 63(4), 1471–1525. Code: github.com/sischei/DeepEquilibriumNets. [link]

  The flagship paper the DEQN lectures are built on: neural networks trained unsupervised to satisfy equilibrium conditions along simulated paths, with heterogeneity, uncertainty, and occasionally binding constraints.

Deep Surrogates to Estimate Dynamic Models (afternoon)

- Chen, Didisheim & Scheidegger (2026). *Deep Surrogates for Finance: With an Application to Option Pricing.* Journal of Financial Economics 177, 104222. [link]

Neural-network surrogates approximate expensive structural models to accelerate estimation by orders of magnitude, here applied to a high-dimensional option pricing model.

Heterogeneous agent models: DeepHAM and structural reinforcement learning (afternoon)

- Han, Yang & E (2026). *DeepHAM: A Global Solution Method for Heterogeneous Agent Models with Aggregate Shocks.* Quantitative Economics 17(2), 297–341. Code: github.com/frankhan91/DeepHAM. [link]

  Neural networks approximate value and policy functions and represent the cross-sectional distribution via learned generalized moments. Focus on the moment representation and the simulation-based training objective.

- Yang, Wang, Schaab & Moll (2025). *Structural Reinforcement Learning for Heterogeneous Agent Macroeconomics.* arXiv:2512.18892. [link]

  A structural RL method that uses low-dimensional prices as state variables and lets agents learn equilibrium price dynamics from simulated paths (Krusell–Smith, Huggett with aggregate shocks, HANK).

Recommended background

- Scheidegger (2026). *Deep Learning for Solving and Estimating Dynamic Models in Economics and Finance.* Textbook treatment, arXiv:2605.14493. Code: github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models. [link]

  Book-length treatment of the whole first half of the school, from the basics of deep learning through Deep Equilibrium Nets to deep surrogates for estimation. The natural reference to consult alongside the lectures, and the source of the hands-on exercises.

- Aiyagari (1994). *Uninsured Idiosyncratic Risk and Aggregate Saving.* Quarterly Journal of Economics 109(3), 659–684. [link]

- Krusell & Smith (1998). *Income and Wealth Heterogeneity in the Macroeconomy.* Journal of Political Economy 106(5), 867–896. [link]

  The classic heterogeneous-agent model with aggregate risk; its curse of dimensionality is exactly what the deep-learning solvers are designed to overcome.

- Judd (1998). *Numerical Methods in Economics.* MIT Press. [link]

- Sutton & Barto (2018). *Reinforcement Learning: An Introduction.* 2nd ed., MIT Press (full text free online). [link]

Day 2 — PDEs, Continuous Time, and the Foundations of Sequence Modelling

Simon Scheidegger (University of Lausanne), Yucheng Yang (University of Zurich), Markus Leippold (University of Zurich)

Physics-informed neural networks for solving partial differential equations (morning)

- Raissi, Perdikaris & Karniadakis (2019). *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations*. Journal of Computational Physics 378, 686–707. [link]

  The paper that started the PINN literature. Read Sections 1–3 carefully; the residual-based loss it introduces is the workhorse of the entire continuous-time toolkit.

Deep learning for continuous-time models (late morning)

- Achdou, Han, Lasry, Lions & Moll (2022). *Income and Wealth Distribution in Macroeconomics: A Continuous-Time Approach*. Review of Economic Studies 89(1), 45–86. [link]

  Recasts the Aiyagari–Bewley–Huggett model in continuous time as coupled HJB and Kolmogorov-forward PDEs, the continuous-time heterogeneous-agent toolkit underlying this session.

- Gu, Laurière, Merkel & Payne (2024). *Global Solutions to Master Equations for Continuous Time Heterogeneous Agent Macroeconomic Models*. arXiv:2406.13726. [link]

  Introduces Economic-Model-Informed Neural Networks (EMINNs) to compute global solutions of master equations, with the continuous-time Krusell–Smith model as the lead application.

- Payne, Rebei & Yang (2025). *Deep Learning for Search and Matching Models*. Swiss Finance Institute Research Paper No. 25-05. [link]

  Extends the deep-learning-for-PDE approach to labor-market models with aggregate shocks and two-sided heterogeneity, and estimates them by simulated method of moments.

Foundations of Sequence Modelling: From RNNs to Transformers (afternoon)

- Hochreiter & Schmidhuber (1997). *Long Short-Term Memory*. Neural Computation 9(8), 1735–1780. [link]

  The gated recurrent architecture that made long-range dependence learnable, and the baseline against which attention is motivated.

- Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser & Polosukhin (2017). *Attention Is All You Need*. Advances in Neural Information Processing Systems 30. [link]

  The Transformer. Everything on Day 3 is a variation on this architecture, so read it before the LLM sessions.

Further readings for this session will be added by the lecturer before the school.

Day 3 — Large Language Models and Deep Sequence Modelling in Finance

Markus Leippold (University of Zurich), Fabio Trojani (University of Geneva)

The LLM Revolution: Scaling Laws and Modern Architectures (morning)

- Devlin, Chang, Lee & Toutanova (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. Proceedings of NAACL-HLT 2019, 4171–4186. [link]

- Brown, Mann, Ryder et al. (2020). *Language Models are Few-Shot Learners.* Advances in Neural Information Processing Systems 33. [link]

  The GPT-3 paper: the counterpart to BERT's encoder paradigm, and the origin of in-context learning.

- Kaplan, McCandlish, Henighan et al. (2020). *Scaling Laws for Neural Language Models.* arXiv:2001.08361. [link]

  Read alongside Hoffmann et al. (2022), "Training Compute-Optimal Large Language Models" (Chinchilla), which revises the compute-optimal trade-off between model size and data.

Further readings for this session will be added by the lecturer before the school.

Domain Adaptation: Building Financial & Climate Intelligence (late morning)

- Hu, Shen, Wallis, Allen-Zhu, Li, Wang, Wang & Chen (2022). *LoRA: Low-Rank Adaptation of Large Language Models.* International Conference on Learning Representations (ICLR). [link]

  The parameter-efficient fine-tuning method used in the hands-on part of this session.

- Webersinke, Kraus, Bingler & Leippold (2022). *ClimateBert: A Pretrained Language Model for Climate-Related Text.* Proceedings of the AAAI 2022 Fall Symposium, arXiv:2110.12010. Models: huggingface.co/climatebert. [link]

  The instructor's own domain-adaptation case study: further pretraining on two million climate-related paragraphs, and what it buys on downstream classification tasks.

- Ouyang, Wu, Jiang et al. (2022). *Training language models to follow instructions with human feedback.* Advances in Neural Information Processing Systems 35. [link]

  The InstructGPT/RLHF paper. Pair it with Rafailov et al. (2023), "Direct Preference Optimization", the simpler alignment objective that has largely replaced it in practice.

Further readings for this session will be added by the lecturer before the school.

Advanced Frontiers: From RAG to Reasoning and Agentic LLMs (afternoon)

- Lewis, Perez, Piktus et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.* Advances in Neural Information Processing Systems 33. [link]

- Vaghefi, Stammbach, Muccione et al. (2023). *ChatClimate: Grounding conversational AI in climate science.* Communications Earth & Environment 4, 480. [link]

  RAG applied end to end in a domain setting: GPT-4 grounded in the IPCC AR6 report, with the answers evaluated by IPCC authors. The template for the applied part of this session.

- Yao, Zhao, Yu, Du, Shafran, Narasimhan & Cao (2023). *ReAct: Synergizing Reasoning and Acting in Language Models.* International Conference on Learning Representations (ICLR). [link]

  Interleaved reasoning traces and tool calls, the basic loop behind agentic LLMs. Read after Wei et al. (2022), "Chain-of-Thought Prompting".

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Using Deep Sequence Modelling in Finance (afternoon)

- Gaegauf, Scheidegger & Trojani (2023). *A Comprehensive Machine Learning Framework for Dynamic Portfolio Choice With Transaction Costs.* Swiss Finance Institute Research Paper No. 23-114. [link]

  The instructor's own framework: Gaussian-process regression plus Bayesian active learning inside a dynamic programming algorithm, which characterizes the irregular no-trade region and scales to many risky assets.

- Gu, Kelly & Xiu (2020). *Empirical Asset Pricing via Machine Learning.* Review of Financial Studies 33(5), 2223–2273. [link]

  The benchmark comparison of machine-learning methods for return prediction, and the reference point for what deep sequence models must beat.

- Leippold (2023). *Sentiment spin: Attacking financial sentiment with GPT-3.* Finance Research Letters 55, Part B. [link]

  A cautionary counterpart to the LLM sessions: dictionary-based financial sentiment measures can be adversarially manipulated with close to 99% success, while context-aware models remain robust.

Further readings for this session will be added by the lecturer before the school.

Recommended background

- Gentzkow, Kelly & Taddy (2019). *Text as Data.* Journal of Economic Literature 57(3), 535–574. [link]

  The standard survey of how economists turn text into data, and the natural bridge from Day 3's methods to empirical work in economics and finance.

- Chen, Pelger & Zhu (2024). *Deep Learning in Asset Pricing.* Management Science 70(2), 714–750. [link]